Problem Set 3

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PY 451 Section A1

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1 Representations and Basis States

Even though our space is infinite-dimensional, we can isolate v_1, v_2 to consider a 2-dimensional subspace. The most general possible choice in this subspace, as illustrated by equation (1.1), is

$$|1\rangle = \cos\theta e^{i\phi_1} v_1 + \sin\theta e^{i\phi_2} v_2,$$

$$|2\rangle = \sin\theta e^{i\phi_3} v_1 - \cos\theta e^{i(\phi_3+\phi_1-\phi_2)} v_2.$$

Idfk what to do for b

2 Operators

1. h

2. Note that $h(x)g(x) = 0$ (except possibly on $\{1/3, 2/3\}$ depending on how the drawing is interpreted, but this set has lebesgue measure 0 so the following conclusion still holds). It follows that

$$\int_0^1 h(y)g(y)g(x)\,\mathrm{d}y = 0.$$

Therefore,

$$(Ah)(x) = \int_0^1 h(y)^2 h(x)\,\mathrm{d}y + \int_0^1 h(y)v(y)v(x)\,\mathrm{d}x.$$

Note that $v(x) = \frac{1}{3}h(x)$. It follows that

$$(Ah)(x) = \int_0^1 h(y)^2 h(x)\,\mathrm{d}y + \int_0^1 \frac{1}{9} h(y)^2 h(x)\,\mathrm{d}y = \frac{10}{9} h(x) \int_0^1 h(y)^2\,\mathrm{d}y.$$

Also note that since $0^2 = 0$ and $1^2 = 1$, we have that $h(x)^2 = h(x)$, so

$$(Ah)(x) = \frac{10}{9} h(x) \int_0^1 h(y)\,\mathrm{d}y = \frac{20}{27} h(x).$$

3. bruh